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Assignment - 1

CSE A & B

23CSE113 User Interface Design

Answer all questions

1. Write a detailed report on Electron framework. Explain its architecture, key features, and its significance in cross-platform desktop application development. Include real-world examples of applications built with Electron.
2. Explain how object communication is handled between the main process and the renderer process in Electron using IPC (Inter-Process Communication). Include a code example demonstrating the communication between a button click (in HTML) and the main process (in JavaScript).
3. Compare and contrast JavaFX, WPF, and GTK in terms of:
    - Language support
    - Platform compatibility
    - UI design capabilities
    - Ease of use

    Based on your research, which framework would you recommend for a cross-platform application and why?
4. Select any open-source Electron application (e.g., Visual Studio Code, Slack, or Postman) and analyze its user interface design. Discuss what UI/UX principles are followed, what can be improved, and what makes the interface user-friendly or not.
5. Write a report comparing how Electron, JavaFX**,** and GTK handle cross-platform deployment. Include details like:
    - Packaging
    - Dependencies
    - OS-specific issues
    - Developer experience

6. Research and analyze the performance and memory usage of applications built using Electron vs. traditional frameworks (like WPF or JavaFX). Why is Electron often criticized for performance? Suggest ways to optimize Electron apps.